

Mehedi Hasan

✉ mmehedihasan29088@gmail.com 🌐 My Portfolio 🔄 mehedihasan88 🌐 mehedi-hasan88 📍 Dhaka, Bangladesh

EDUCATION

Jahangirnagar University
MSc. in Computer Science & Engineering

Aug 2023 – Sep 2024
GPA: 3.50/4.0

Jahangirnagar University
BSc. in Computer Science & Engineering

Feb 2018 – July 2023
GPA: 3.47/4.0

EXPERIENCE

Samsung R&D Institute Bangladesh
Senior Software Engineer

Dhaka, Bangladesh
Jan 2024 – Present

Innovation & Research | Key Field: ML, NLP, Agentic AI, XAI, Wearable devices

- **Developed an interactive prior-art search agent that refines user queries and detects multi-patent obviousness relationships with explanations;**
- Fine-tuned an on-device model for Samsung Galaxy Watch for contextual positioning of the floating ongoing icon on the watchface, improving detection accuracy by 21% over the base model.
- Proposed initial idea and contributed to the development of a **novel machine learning-based methodology for liquid consumption estimation using wearable devices.**
- Developed a weekly report summarizer from GitHub commit messages.
- Fine-tuned Bangla NER model and deployed on a mobile device.
- Actively contribute to team innovation meetings by evaluating peer-submitted ideas, providing constructive technical feedback on novelty, obviousness, and feasibility.

Development | Tech Stack: JAVA, Kotlin, Jetpack Compose, Android, HILT, WearOS

- Developed core modules (notifications, nowbar, watchface overlay, BnR) for Samsung Galaxy Watch 7/8, optimizing performance and performing feature prototyping leading junior engineers.
- Designed and built backend architecture for Nowbar using concurrent data structures; integrated AOD overlay system and improved latency from 30ms to 5ms via MVVM migration.
- Developed a POC for a link-preview system in the watch notification panel, identifying security vulnerabilities and proposing corresponding solutions.

PUBLICATIONS

- **Mehedi Hasan, Mahbub Islam Mahim (2026). CTPG-RAG: Causal-Temporal Patient Graph Retrieval-Augmented Generation for Evidence-Grounded Clinical Outcome Attribution | Work In Progress**
 - Standard RAG flattens EHR data into text, ignoring temporal and clinical relationships and thus losing critical event dependencies. To address this limitation, we propose CTPG-RAG, a framework that models EHR into a directed graph and enables backward tracing of ICU outcomes to clinically grounded causes.
- **Mehedi Hasan, Mahbub Islam Mahim, Jugal Krishna Das (2026). Segmentation Strategies for Long-Code Authorship Detection: Balancing Context Coverage and Computational Efficiency. | Accepted**
 - Authorship detection degrades when long code exceeds transformer context window, losing stylistic and structural cues. We have compared truncation, multi-view, and sliding-window strategies to handle context overflow, highlighting trade-offs in performance and efficiency.
- **Mehedi Hasan, Mahbub Islam Mahim (2025). Divide and Conquer: A Two-Stage Label-Wise Cascaded Framework for Bangla Hate Speech Detection. | Codes🔗 Paper🔗**
 - Proposed a cascaded framework combining binary and multiclass models with k-fold cross-validation and intra-label ensembling to tackle low-resource hate speech classification.
 - Received **Best Paper Award**🔗 at the BLP Workshop 2025, and presented at AACL-IJCNLP 2025.

SELECTED PROJECTS

- **Layerwise-Probes-Ablations-in-BERT** – *PyTorch, Hugging Face Transformers, scikit-learn, Matplotlib*
 - Analyzed BERT’s 12-layer encoder using linear probing and layer ablation to distinguish layer-wise responsibilities. Demonstrated that mid-layers (5–6) yield the most separable features, while upper layers (10–11) drive final predictions in parts-of-speech tagging and semantic similarity tasks.
- **Human-Machine-Hybrid-Adversarial-Code-Classification** – *PyTorch, scikit-learn, NumPy, Pandas*
 - Investigated multiple approaches to multilingual code classification — including AST-based feature engineering, overlapping window analysis, multi-view learning, and specialized training loops
- **Machine Learning based Image Forgery Detection** – *OpenCV, NumPy, SciPy, scikit-learn, Matplotlib, Seaborn*
 - Developed an ensemble-based image forgery detection pipeline combining SVM, Random Forest, and Gradient Boosting on LBP-DCT features, outperforming the individual classifiers.

COMPETITIONS & ACHIEVEMENTS

Competitive programming & Problems solving

- Solved **1,900+ algorithmic problems** across multiple online judges, participated at **160+ contests**.
- Achieved highest rating of **1611 (Expert)** on Codeforces (@MhD88) and **1730 (3-star)** on CodeChef (@MhD88).
- Top Contest Ranks: Top 2% among 12,275 participants (@MhD88), Top 2% among 16,010 participants (@MhD88)

Hackathons & ML Contests

- Became **Finalist among 400+** teams (Team: JU_NONAME) at Code Samurai Hackathon 2022, hosted by University of Dhaka. Solved algorithmic and SQL problems in preliminaries; developed a web-based system in the final.
- Ranked **1st among 90 teams** (Team: team_hate_mate) at the BLP Workshop 2025 Shared Task

SCHOLARSHIPS & AWARDS

- Received **Excellence Award** twice for contribution in innovation and development at Samsung during Apr-Jun 2025 and Jan-Mar 2024
- Received an award for achieving the professional-level Software Coding (SWC) certification test on November 08, 2024, in recognition of coding excellence.
- Received **Dean’s Merit scholarships** in all four years of B.Sc. for academic excellence (2018–2023)
- Received prestigious national merit scholarship in both primary and secondary school certification examination (2011, 2013)

TEACHING EXPERIENCES

Competitive Programming Trainer | *Department of CSE, Jahangirnagar University* 2022 - 2023

- Provided training on C/C++, Data Structures & Algorithms, problem solving strategies
- Organized weekly programming contests; and coordinated annual sessions to attract new students to community.

Academic Tutoring | *Private Tutor* 2020 - 2024

- Provided academic mentorship to over 80 students across high school, college, and university levels in physics, mathematics, and programming disciplines.

SKILLS

Programming Languages	: C, C++, Java, Kotlin, Python, JavaScript
ML & AI	: scikit-learn, NumPy, OpenCV, PyTorch, Hugging Face Transformers, LLMs
Frameworks & Technologies	: Android, Node.js, Jetpack Compose, HILT, Bootstrap
Databases	: MySQL, MongoDB
Tools	: Git, Perforce, VS Code, Android Studio